



Installing JetPack SDK on Jetson

Setting up the foundation for Edge AI development

A comprehensive guide for developers and engineers to get your Jetson device ready for Edge AI applications

What Is JetPack SDK?

NVIDIA's **all-in-one software stack** for Jetson devices that provides everything needed for [AI at the edge](#)



Includes:

Ubuntu-based OS image

CUDA, cuDNN, TensorRT

DeepStream SDK

Multimedia APIs & drivers



Why JetPack Matters

Optimized for Jetson Hardware

Custom-tuned for NVIDIA's edge computing platform to maximize efficiency

Maximum GPU Performance

Unlocks the full potential of Jetson's GPU cores for AI workloads

Simplified Development

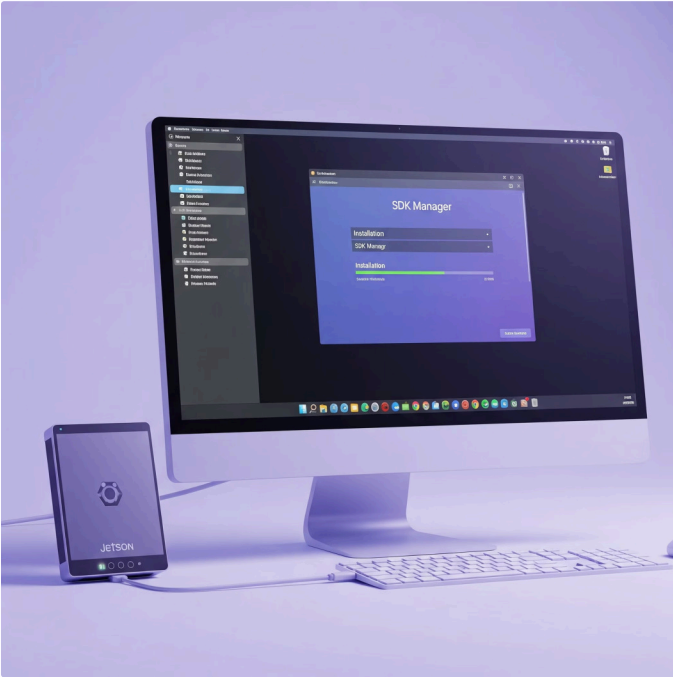
Streamlines AI, computer vision, and robotics development with pre-optimized libraries

Regular Updates

Stays in sync with the broader CUDA ecosystem for latest features and optimizations

Installation Options

Option 1: NVIDIA SDK Manager



- Flash Jetson device with complete JetPack image
- Install additional components post-flash
- Hosted on Ubuntu PC

Option 2: Pre-flashed Developer Kits



- Comes with JetPack preinstalled
- Update via apt package manager
- Or use SDK Manager for major version upgrades

Step 1: Prepare the Jetson Device



Hardware Options

Jetson Nano, Xavier, or Orin platform based on your performance needs



Required Peripherals

HDMI monitor, USB keyboard/mouse, microSD/eMMC storage



Power Requirements

5V/4A or higher power supply (varies by model)



Network Connection

Ethernet cable or compatible Wi-Fi adapter for updates

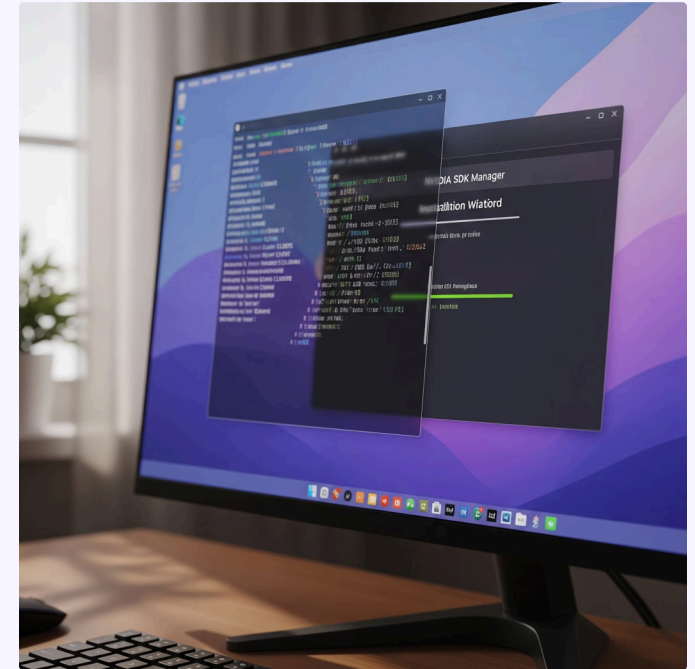
Step 2: Install NVIDIA SDK Manager

On your Ubuntu host PC:

```
sudo apt update  
wget  
https://developer.download.nvidia.com/sdkmanager/sdkmanager\_latest.de  
b  
sudo apt install ./sdkmanager_latest.deb
```

Launch with: `sdkmanager`

 **Important:** SDK Manager requires Ubuntu 18.04 or newer on the host PC. It will not work on other Linux distributions or operating systems.



SDK Manager provides a GUI interface for managing all aspects of JetPack installation and updates.

Step 3: Flash Jetson with JetPack

Connect Hardware

Connect Jetson to host PC via micro-USB (Nano/NX) or USB-C (Xavier/Orin) cable



SDK Manager Setup

Select your Jetson model and desired JetPack version

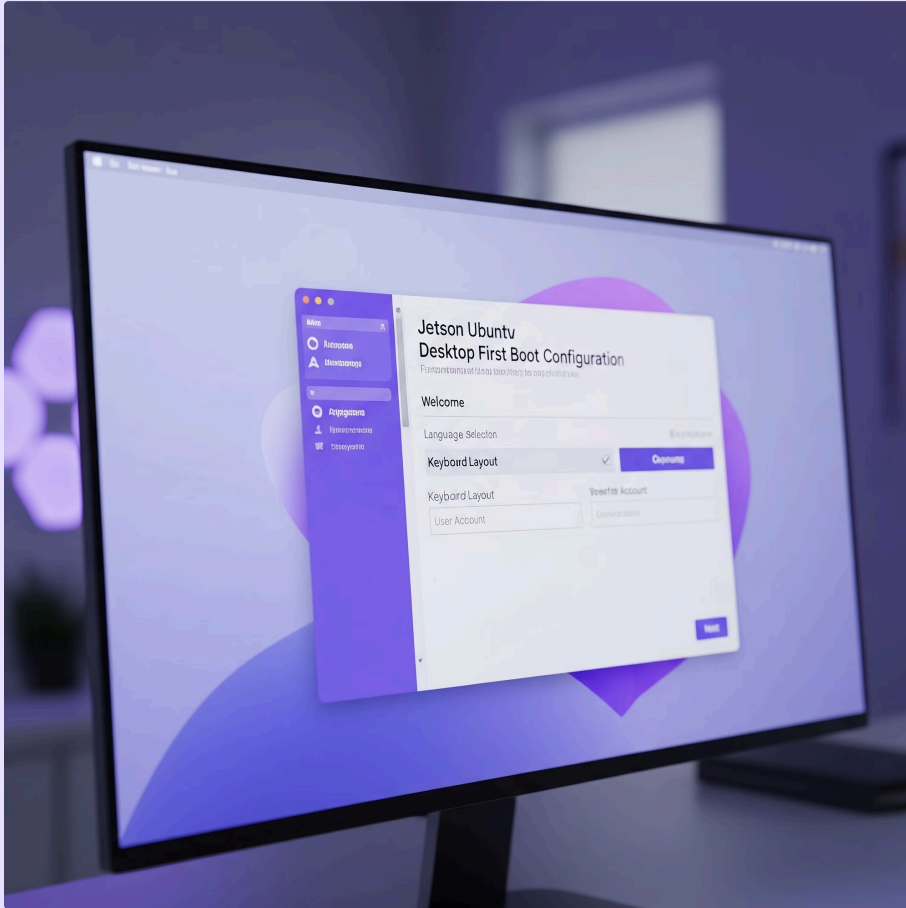
Recovery Mode

Put Jetson into [recovery mode](#) by holding recovery button while powering on

Begin Flashing

Start installation to flash OS and install CUDA, cuDNN, and TensorRT packages

Step 4: First Boot Setup



Initial Configuration

Set username and password credentials

Configure network settings (Wi-Fi or Ethernet)

Complete system updates:

```
sudo apt update && sudo apt upgrade -y
```

Accept license agreements when prompted

Step 5: Verify JetPack Installation

Check CUDA Installation

```
nvcc --version
```

Should show CUDA compiler version matching your JetPack

Verify TensorRT

```
python3 -c "import tensorrt; print(tensorrt.__version__)"
```

Confirms TensorRT libraries are properly installed

Run Sample Applications



Test with DeepStream demos or camera capture examples to ensure all components are working properly



Post-Install Extras



AI Frameworks

Install [PyTorch](#) and [TensorFlow](#) Jetson-optimized builds from NVIDIA's repositories



Jetson.GPIO

Enable GPIO library for connecting sensors and IoT components to your Jetson device



Performance Tuning

Optimize swap space and power settings for handling heavy AI model inference



Jetson Zoo

Access community repository for additional AI libraries and optimized packages

Common Issues & Solutions

Power Problems

Issue: Random shutdowns or unstable performance

Cause: Inadequate power supply (under-rated adapter)

Solution: Use NVIDIA-recommended power adapter (5V/4A for Nano)

Storage Limitations

Issue: Installation fails at OS flashing stage

Cause: Insufficient SD card/eMMC space

Solution: Use 32GB+ Class 10 microSD or larger eMMC module

Software Compatibility

Issue: Flashing errors or SDK Manager crashes

Cause: Outdated SDK Manager or incompatible host OS

Solution: Update SDK Manager and check NVIDIA compatibility docs

Always check NVIDIA documentation and Jetson forums for latest troubleshooting guidance



Key Takeaways

1

Foundation

JetPack SDK is the **core software foundation** for all Jetson AI projects

2

Installation

Install via **SDK Manager** for full control or use pre-flashed images for convenience

3

Tools

Provides **CUDA, cuDNN, TensorRT, and DeepStream** out of the box

4

Verification

Always verify with test scripts before deploying real models in production